

ML Project : Diabetes Patient Readmission Prediction Using Machine Learning

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DIA - 4

1 Business scope and dataset

Hospital readmission among diabetic patients is a major concern for healthcare systems, as it affects both quality of care and hospital resource management. The objective of this project is to develop a machine learning model capable of predicting whether a patient will be readmitted within 30 days, after 30 days, or not at all.

This problem is particularly relevant to our field of specialization in Data and Artificial Intelligence. Our goal is to understand how advanced data-driven methods can be applied to real-world medical challenges, and how predictive models can support clinical decision-making. Working on hospital readmission allows us to explore the concrete impact that AI techniques can have in healthcare, a domain where improving patient outcomes and optimizing resources are critical issues.

The dataset used in this project is publicly available on Kaggle [1], and contains detailed demographic, diagnostic, and hospitalization information for diabetic patients.

Such a model can support hospitals by:

- identifying high-risk patients at discharge,
- optimizing resource allocation (monitoring, follow-up consultations),
- reducing preventable readmissions,
- improving patient management and care pathways.

Although the dataset used contains records up to 2008, the prediction task remains clinically and operationally relevant today. The variables included in the dataset: age, comorbidities, medication patterns, length of stay, prior hospital visits; are still used in contemporary hospital decision-making processes. Moreover, obtaining recent medical datasets is particularly challenging due to strict privacy regulations, long anonymization procedures, and limited public availability of electronic health records. As a result, many studies in healthcare machine learning rely on historical datasets. Therefore, the insights and models developed here remain applicable to modern healthcare contexts despite the age of the data.

2 Problem formalisation and methods

The task is formulated as a multiclass classification problem:

- **0** = No readmission,
- **1** = Readmission after 30 days,
- **2** = Readmission within 30 days.

The dataset presents several challenges:

- strong class imbalance, particularly for the “<30 days” class,

- many categorical features (race, medication types, ICD-9 codes),
- noisy or inconsistent medical data entries,
- high-dimensional sparse features after one-hot encoding.

Several families of models were evaluated:

- baseline models: Logistic Regression, Decision Tree, KNN, Random Forest, XGBoost
- class imbalance strategies: class weights, SMOTE (tested but not retained),
- dimensionality reduction: PCA (no performance gain),
- ensemble learning: Bagging, Soft Voting, Stacking,
- advanced model: CatBoost, chosen based on literature demonstrating strong performance on tabular data with many categorical variables.

The main evaluation metric is the **F1-macro score**, which treats each class equally and is more appropriate than accuracy for imbalanced datasets.

2.1 Algorithm description

Baseline models were constructed first to obtain reference performances. Hyperparameter tuning was conducted using RandomizedSearchCV.

Finally, CatBoost was tested as a state-of-the-art algorithm for categorical data. Literature such as Dorogush et al. (2018) shows that CatBoost handles categorical embeddings, missing values, and high-dimensional tabular structures efficiently, making it a strong candidate for this problem.

2.2 Limitations

The project faces several limitations:

- severe class imbalance despite class weighting strategies,
- sparse high-dimensional feature space after one-hot encoding,
- limited predictive performance due to noisy and heterogeneous real-world medical data.

3 Methodology

3.1 Data description and exploration

The dataset includes demographic features, diagnoses (ICD-9 codes), medication information, hospitalization details, and prior visit history. Because ICD-9 codes are highly granular and medically specific, we consulted an external reference [2] to better interpret them. Exploratory analysis highlighted several important patterns in the population.

- an overrepresentation of older patients,
- a heterogeneous distribution of diagnoses,
- frequent use of diabetes-related medications (insulin),
- a heavy dominance of the “no readmission” class.

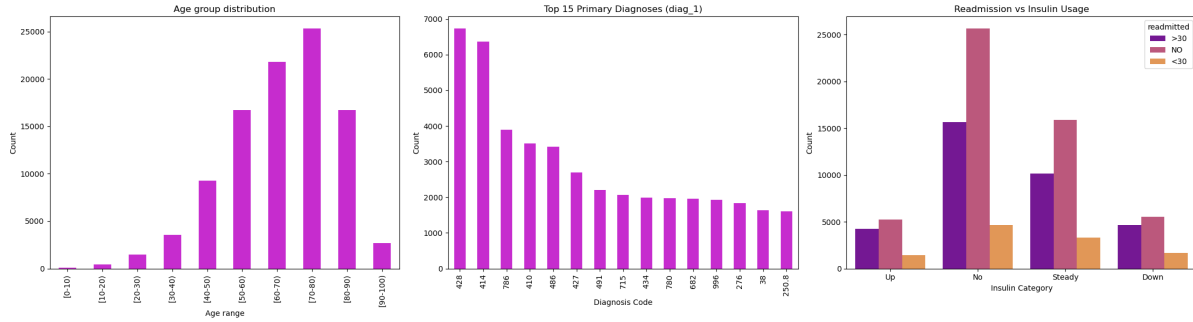


Figure 1: Overview of key variables: age distribution, primary ICD-9 diagnoses, and insulin usage by readmission category.

3.1.1 Missing values

During the exploratory phase, we identified several inconsistencies and missing data issues in the dataset. Some columns contained the character “?” instead of proper missing values, so these entries were first converted to `NaN` to ensure consistent processing.

A few variables presented a very high proportion of missing values. In particular, the columns `weight`, `payer_code`, and `medical_specialty` contained more than 50% missing entries and were therefore removed from the dataset.

We also noticed that the `gender` column included three unique values: `Male`, `Female`, and `Unknown/Invalid`. Since the third category is not clinically meaningful and represents a very small number of cases, all rows with this label were removed.

For the variables `diag1`, `diag2`, `diag3`, and `race`, the proportion of missing values was very small compared to the dataset size. These rows were therefore dropped without significantly impacting the dataset.

Finally, the columns `max_glu_serum` and `A1Cresult` contained the label `None`, which does not indicate a missing value but rather that the laboratory test was not performed. Since these values were incorrectly interpreted as missing by default, this issue was corrected during preprocessing to preserve the clinical meaning of the variables.

3.1.2 Imbalanced data

The dataset is strongly imbalanced:

- the majority class (no readmission) dominates,
- the minority class (readmitted within 30 days) is particularly small.

Solutions explored included:

- class weights in the loss function,
- SMOTE oversampling (tested but not retained due to average results).

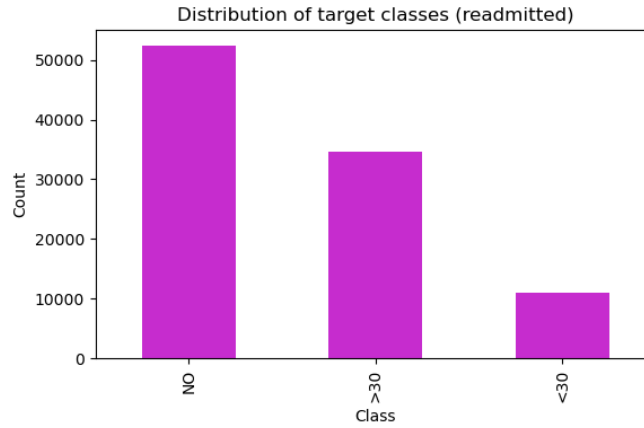


Figure 2: Distribution of target classes (readmitted)

3.1.3 Outliers

We do not perform an outlier analysis because most features are categorical, and the others are medical measures (e.g., glucose, BMI) whose extreme values may represent real medical conditions rather than data errors.

3.2 Data splitting for train/test

Before splitting the dataset, two cleaned versions were created: a non-scaled dataframe (`df_clean`) and a scaled version (`df_scaled`). Both were saved to the `data/` directory to ensure full reproducibility of the preprocessing pipeline.

For each version, the features and target were then separated and split into training and testing sets. This resulted in four datasets: `X_train`, `X_test`, `y_train`, and `y_test` for the non-scaled data, as well as `X_train_scaled` and `X_test_scaled` for the scaled version. All splits were exported as CSV files and later reloaded in the modeling notebook for training and evaluation:

- `df_clean.csv`, `df_scaled.csv`
- `X_train.csv`, `X_test.csv`
- `y_train.csv`, `y_test.csv`
- `X_train_scaled.csv`, `X_test_scaled.csv`

3.3 Algorithm implementation and hyperparameters

Hyperparameter tuning was performed using `RandomizedSearchCV` for the following models:

- Logistic Regression,
- Decision Tree,
- Random Forest,
- XGBoost.

We chose `RandomizedSearchCV` instead of a traditional Grid Search approach because several of our models, particularly Random Forest and XGBoost, contain a large number of hyperparameters. Performing an exhaustive Grid Search would require evaluating hundreds or even thousands of parameter combinations, which is computationally expensive, especially when combined with 5-fold stratified cross-validation.

Randomized Search provides a more efficient alternative by sampling a limited number of random hyperparameter combinations across a wide search space. In practice, this method often reaches similar or even superior performance compared to Grid Search while significantly reducing computation time. This makes it a more realistic and scalable choice for our project.

In addition to hyperparameter tuning, we implemented several ensemble learning strategies, including bagging, soft voting, and stacking. Finally, CatBoost was evaluated due to its strong performance on tabular data and its native handling of categorical variables.

4 Results

4.1 Metrics

Because the majority of patients are **not readmitted**, a naive model that always predicts “no readmission” would still achieve a relatively high accuracy. For this reason, plain accuracy is not a reliable metric in our context. Instead, we evaluate our models using metrics that better reflect performance on the minority classes.

In this project, we focus on the following metrics:

- **F1-macro**

This is our main evaluation metric. It computes the F1-score independently for each class and then averages them, giving equal importance to minority and majority classes. This is particularly appropriate since both readmission classes are underrepresented.

- **F1-weighted**

Similar to F1-macro, but each class F1-score is weighted by its number of samples. This metric provides a complementary view by reflecting the influence of the dominant “no readmission” class.

- **Recall**

Recall measures the proportion of true positive cases correctly identified by the model. In a medical setting, missing a patient at high risk of readmission can be costly; therefore, recall is especially important for the readmitted classes.

- **ROC-AUC (macro)**

The ROC-AUC score evaluates the model’s ability to discriminate between classes across different decision thresholds. We report the macro-averaged ROC-AUC to summarize the discriminative performance over all three classes.

4.2 Evaluation and comparison with the baseline

Across all experiments, the best overall performance was obtained with the tuned Random Forest model, achieving approximately:

$$\text{F1-macro} \approx 0.45.$$

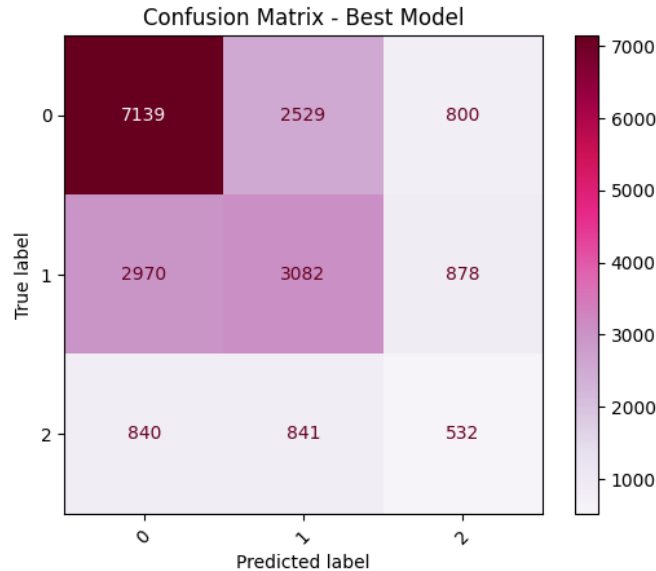


Figure 3: Confusion matrix of our best model : Tuned Random Forest

This performance is consistent with results reported in the literature for this same dataset, due to its complexity and class imbalance. Ensemble models such as stacking were competitive but did not surpass the tuned Random Forest.

CatBoost produced stable performance and confirmed its suitability for tabular data, but did not significantly outperform Random Forest in this case.

5 Working Methodology

This project was carried out collaboratively by two team members. To organize our workflow efficiently, we created a shared GitHub repository and adopted a branching strategy: each of us worked on our own branch, then merged our contributions into the `dev` branch before integrating the final validated version into `main`. This ensured clean version control and reduced conflicts during development.

Although GitHub records individual commits, the commit counts and contribution percentages do not accurately reflect the real distribution of work. We made the decision to perform most project steps together, working side by side on a single computer during our meetings, sometimes on Sandrine's laptop, sometimes on Ovia's. This collaborative approach allowed us to discuss preprocessing choices, modeling strategies, and design decisions in real time, ensuring that every part of the project was developed jointly.

For transparency, the structure of the repository and the organization of all files can be found in the project's README file on GitHub.

6 Discussion and Conclusion

This project set out to predict hospital readmission among diabetic patients, addressing our business objective of exploring how Data and AI can support clinical decision-making. Through data cleaning, handling missing values, managing class imbalance, and training several machine learning models with hyperparameter tuning, we built a pipeline capable of extracting meaningful patterns from complex hospital data.

Random Forest achieved the best overall performance, with an F1-macro score around 0.45. While this level of performance is not sufficient for operational deployment in a clinical setting, it still provides informative signals about the factors associated with readmission and illustrates the potential of predictive modelling in healthcare. The moderate performance mainly reflects the difficulty of the task, the high class imbalance, and the noise inherent in real-world medical datasets.

Future improvements could include richer feature engineering, temporal modelling, or the use of larger and more recent datasets. Overall, the project demonstrates how Data and AI techniques can be applied to a real medical problem and highlights both the opportunities and limitations of predictive analytics in healthcare.

References

- [1] Brandão, D. *Diabetes 130-US hospitals for years 1999–2008*. Available on Kaggle: (copy paste the link) https://www.kaggle.com/datasets/brandao/diabetes?select=diabetic_data.csv
- [2] Government of British Columbia. *Diagnostic Code Descriptions (ICD-9)*. Available at: <https://www2.gov.bc.ca/gov/content/health/practitioner-professional-resources/msp/physicians/diagnostic-code-descriptions-icd-9>
- [3] UCI Machine Learning Repository. *Diabetes 130-US Hospitals for Years 1999–2008*. Available at: <https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>.